

AFRA AZAD

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EDUCATION

Bachelor of Arts and Science, University of Toronto St. George — Data Science
Dean Scholar List — CGPA: 3.91

Sep 2021 to Dec 2026
Later Life Learning OSOTF Award

SKILLS

Technical Skills: LLMs, NLP, Transformer Models (BERT, LLaMA), Ensemble Learning; Python (HuggingFace, Scikit-learn, TensorFlow), Postgres, SQL, SAS; Power BI (DAX), Tableau; Google Cloud, MS Azure; Git, Anaconda

EXPERIENCE

Machine Learning Developer
UHN (University Health Network)

September 2025 – Present
Toronto, ON

- Performed large-scale **data pre-processing** and **data quality validation**, including **unsupervised techniques** such as **outlier detection**, **missing data handling**, **distribution analysis**, **feature engineering** and **PCA-based** dimensionality reduction on 10,000+ patient records.
- Designed and evaluated **supervised learning LLMs** for longitudinal prediction, including **LSTM**, **BiLSTM**, **GRU**, **RNN**, and **1D-CNN** architectures.
- Compared models such as **XGBoost**, **LightGBM**, **GMM**, **Lasso**, and **Ridge**, combined with **stacking** and **blending**.
- Implemented models using **Python** in **SAS**, leveraging **NumPy**, **Pandas**, **Scikit-Learn**, and **TensorFlow/Keras** for end-to-end experimentation and evaluation.
- Conducted extensive **model optimization** and **hyperparameter tuning** with **train-validation split**, evaluating over **5,000 ensemble model configurations** using **RMSE**, **MAE**, **MSE**, **R²**, and **MAPE**.
- Deployed a production-ready **predict model pipeline** extracting training data from **data pipeline** for multiple PROMIS outcome variables, automatically updating results in a **SQL database**.
- Developed final anxiety model prediction report for a non-technical audience for a research study.

Student Data Analyst
University of Toronto Temerty Faculty of Medicine

October 2024 – April 2025
Toronto, ON

- Evaluated existing **NLP Named Entity Recognition (NER)** models using **SpaCy**, **LLaMA**, and **Generative AI** to de-identify **Personally Identifiable Information (PII)** over **33 thousand** for data privacy and security.
- Performed content analysis on **embedded** survey comments using **transformer-based architectures BERT** for feedback over time, utilizing **GitHub**, **Unix**, and **Anaconda** for version control, scripting, and remote computing.
- Applied **transfer learning** by fine-tuning **BERT transformer models** for **sentiment analysis**.
- Implemented **text preprocessing** pipelines including **tokenization**, **feature extraction**, and **data cleaning** for unstructured text data.
- Optimized classification models using **GridSearchCV** and **TF-IDF**, achieving **98.5% accuracy**.

Student Data Analyst Intern
Culligan Quench

June 2023 – August 2023, June 2024 – August 2024
Vaughan, ON

- Applied software engineering to engineer an end-to-end **data pipeline** from a legacy data warehouse (**Watertight**) to **Power BI** using **Python API**.
- Implemented **SQL** queries to support early-stage **data warehouse development** in **MS Azure** database to improve data accessibility for analytics teams.
- Applied **fuzzy matching** and **geospatial analysis** using the **Google Maps API** to join the technician and sales datasets based on **geocoded** addresses.
- Developed automated **Power BI dashboards** using **DAX**, integrating **Microsoft Power Automate** for scheduled refreshes, merging sales and inventory databases for product and customer analysis.
- Clustered 10,000+ customer locations using **unsupervised centroid-based clustering** and optimized delivery zones through **route optimization algorithms**.
- Conducted external **data merging, cleaning and wrangling** of government APIs and commercial datasets (e.g., **Dun & Bradstreet**) with around **hundred thousand** companies to support market analysis.